

CSS452 Internet of Things
Homework 4: ESP32 – MQTT Clients

1. Do the project in Section 4 in Lecture Note 6 as shown in Pages 36 – 51. Take a video to show
 - circuit connection,
 - Arduino code,
 - demonstration:
 - rotate the knob of the potentiometer to turn on the LED,
 - rotate the knob of the potentiometer to turn off the LED,

Name it as Q1. Submit the video to the Google Classroom.

2. Work as a group of two students. We will use **two** ESP32. Let us call them Node1 and Node2.
 - Each ESP32 will work as both publisher node and subscriber node similar to Section 4.
 - Each ESP32 will connect to a potentiometer and an LED similar to Section 4.
 - We rotate the knob of the potentiometer of Node1 will turn on/off the LED of Node2 (the condition is similar to Section 4).
 - We rotate the knob of the potentiometer of Node2 will turn on/off the LED of Node1 (the condition is similar to Section 4).
 - **Hint: Revise the code “Public_HiveMQ_PubSub.ino” and upload to both ESP32.**

Take a video to show

- circuit connection,
- Arduino code,
- demonstration:
 - rotate the knob of the potentiometer of Node1 to turn on/off the LED of Node2,
 - rotate the knob of the potentiometer of Node2 to turn on/off the LED of Node1.

Name it as Q2. Submit the video to the Google Classroom.
